

1. The polyhedron is represented in Figure 1

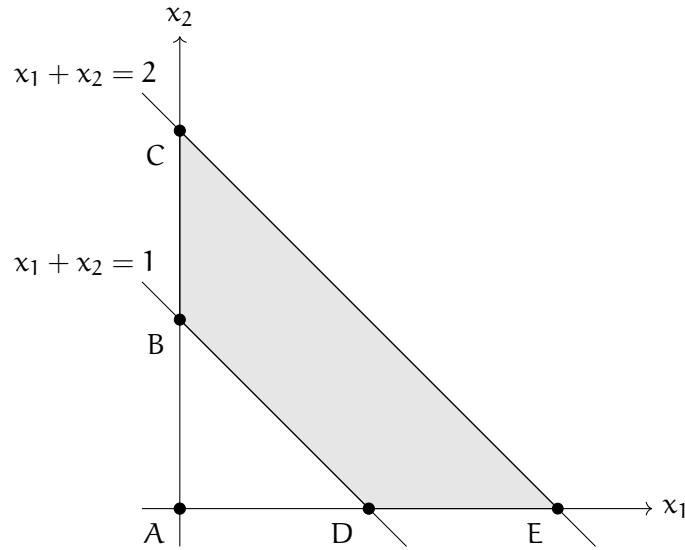


Figure 1: The constraint polyhedron

Note that it has four vertices, labeled B, C, D and E.

2. The polyhedron can be written in standard form by introducing non negative slack variables, and rewriting the constraints in the form of equality constraints:

$$\begin{aligned} -x_1 - x_2 + x_3 &= -1, \\ x_1 + x_2 + x_4 &= 2, \\ x_1, x_2, x_3, x_4 &\geq 0. \end{aligned}$$

The equality constraints in matrix form are

$$Ax = b, \text{ where } A \in \mathbb{R}^{2 \times 4}, b \in \mathbb{R}^2, x \in \mathbb{R}^4,$$

$$A = \begin{pmatrix} -1 & -1 & 1 & 0 \\ 1 & 1 & 0 & 1 \end{pmatrix}, b = \begin{pmatrix} -1 \\ 2 \end{pmatrix}.$$

3. This is a problem with $n = 4$ variables and $m = 2$ equality constraints. Therefore, the total number of potential bases is 6. Let's investigate them.

- (a) x_3 and x_4 in the basis. We consider the third and fourth columns of matrix A which form the matrix

$$B = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

It is invertible, therefore it is a valid basis. We set the non basic variables x_1 and x_2 to zero. Then:

$$x_B = B^{-1}b = \begin{pmatrix} -1 \\ 2 \end{pmatrix}; x = \begin{pmatrix} 0 \\ 0 \\ -1 \\ 2 \end{pmatrix}.$$

x_B has a negative entry. The basic solution is not a *feasible* basic solution. Therefore, it does not correspond to a vertex of the polyhedron. Setting x_1 and x_2 to zero corresponds to point $A = (0, 0)$ in Figure 1. At that point, two constraints are active: $x_1 \geq 0$ and $x_2 \geq 0$, as it is a basic solution. But it is clearly outside of the polyhedron, and it is not a vertex.

- (b) x_1 and x_3 in the basis. We consider the first and third columns of matrix A which form the matrix

$$B = \begin{pmatrix} -1 & 1 \\ 1 & 0 \end{pmatrix}.$$

This matrix is invertible. Therefore, we have a valid basis. We set the non basic variables x_2 and x_4 to zero. Then:

$$B^{-1} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}; x_B = B^{-1}b = \begin{pmatrix} 2 \\ 1 \end{pmatrix}; x = \begin{pmatrix} 2 \\ 0 \\ 1 \\ 0 \end{pmatrix}.$$

x_B is non negative, and we have a feasible basic solution. It corresponds to a vertex of the polyhedron. This is the point $E = (2, 0)$ in Figure 1. At that point, two constraints are active: $x_1 + x_2 \leq 2$ and $x_2 \geq 0$. In standard form, it means that constraints $x_4 \geq 0$ and $x_2 \geq 0$ are active. It shows that setting the non basic variables to zero is equivalent to impose the corresponding constraints to be active.

- (c) x_1 and x_2 in the basis. We consider the first and second columns of matrix A which form the matrix

$$B = \begin{pmatrix} -1 & -1 \\ 1 & 1 \end{pmatrix}.$$

This matrix B is singular. Therefore, this is not a valid basis. From a geometrical point of view, a basis is at the intersection of two constraints. But, here, the non basic variables x_3 and x_4 correspond to two parallel constraints. They have no intersection.

- (d) x_1 and x_4 in the basis. We consider the first and fourth columns of matrix A which form the matrix

$$B = \begin{pmatrix} -1 & 0 \\ 1 & 1 \end{pmatrix}.$$

The basic matrix is invertible. It is therefore a valid basis. We set the non basic variables x_2 and x_3 to zero. Then:

$$B^{-1} = \begin{pmatrix} -1 & 0 \\ 1 & 1 \end{pmatrix}; x_B = B^{-1}b = \begin{pmatrix} 1 \\ 1 \end{pmatrix}; x = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 1 \end{pmatrix}.$$

x_B is non negative, and we have a feasible basic solution. It corresponds to the point $D = (1, 0)$ in Figure 1.

- (e) x_2 and x_3 in the basis. We consider the second and third columns of matrix A which form the matrix

$$B = \begin{pmatrix} -1 & 1 \\ 1 & 0 \end{pmatrix}.$$

It is invertible, and we have a valid basis. We set the non basic variables x_1 and x_4 to zero. Then:

$$B^{-1} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}; x_B = B^{-1}b = \begin{pmatrix} 2 \\ 1 \end{pmatrix}; x = \begin{pmatrix} 0 \\ 2 \\ 1 \\ 0 \end{pmatrix}.$$

It is non negative, and corresponds to the point $C = (0, 2)$ in Figure 1.

- (f) x_2 and x_4 in the basis. We consider the second and fourth columns of matrix A which form the matrix

$$B = \begin{pmatrix} -1 & 0 \\ 1 & 1 \end{pmatrix}.$$

It is invertible, and we have a valid basis. We set the non basic variables x_1 and x_3 to zero. Then:

$$B^{-1} = \begin{pmatrix} -1 & 0 \\ 1 & 1 \end{pmatrix}; x_B = B^{-1}b = \begin{pmatrix} 1 \\ 1 \end{pmatrix}; x = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 1 \end{pmatrix}.$$

x_B is non negative. We have a feasible basic solution. It is a vertex of the polyhedron and corresponds to the point $B = (0, 1)$ in Figure 1.